Empirical Design: College Basketball → Professional Entry

Replicating Nonlinear Peer Effects in Advancement

I. Objective

This document provides a **fully specified empirical design** for testing whether the nonlinear relationship observed in U.S. Army promotions—an inverted-U relationship between peer quality and advancement—replicates in a sports setting.

The goal is to construct a setting where: - individuals are evaluated within **peer groups (teams)**, - advancement is **competitive and capacity-constrained**, and - performance is interpreted **relative to peers**.

II. Conceptual Mapping

Army Setting	College Basketball
Senior rater × time pool	Team × season
Top-block rating	Performance metrics (BPM, efficiency, usage)
Peer quality (mean TB rate)	Average teammate quality
Promotion to Major	Entry into professional basketball
Capacity constraint	Draft slots, roster limits

III. Sample Definition

Population

- NCAA Division I men's basketball players

Time Period

- Recommended: 2000–2023

Unit of Observation

- Player × season panel

Inclusion Criteria

- Minimum playing time (e.g., ≥ 200 minutes)
- Exclude redshirt-only seasons
- Drop observations with missing key stats

IV. Data Sources

1. College Performance Data

- Sports-Reference (College Basketball)
  - https://www.sports-reference.com/cbb/
- Variables:
  - minutes, points, rebounds, assists
  - advanced metrics (BPM, WS, PER)

2. Professional Outcomes

- Basketball-Reference (NBA Draft and player pages)
  - https://www.basketball-reference.com/

3. Matching Strategy

- Name + school + year
- Clean duplicates and spelling variations

V. Variable Construction

A. Outcome Variable

Primary outcome:

$Y_i = 1$ if player enters professional basketball within k years

Operational definitions: - Drafted into NBA (cleanest) - OR plays ≥1 NBA game - Optional: include G-League or major international leagues

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B. Individual Performance

Candidate measures: - Box Plus-Minus (BPM) - Points per minute - Usage-adjusted scoring - Win shares per 40 minutes

Goal: Capture performance that evaluators plausibly observe.

C. Pool Definition

$Pool_{jt}$ = Team j × Season t

Optional: - Coach × season

D. Pool Quality (Key Independent Variable)

Leave-one-out mean:

$PoolQuality_{jt(-i)}$ = average performance of teammates excluding player i

Rationale: Avoid mechanical correlation between individual and pool measure.

E. Control Variables

- Year fixed effects
- Position (guard/forward/center)
- Class year (freshman, sophomore, etc.)
- Team success (win percentage)
- Conference fixed effects (optional)

VI. Empirical Specification

Baseline Model

$Pr(Y_i = 1) = f(\beta_1 Performance_i + \beta_2 PoolQuality_{jt} + \beta_3 PoolQuality_{jt}^2 + Controls + \varepsilon_i)$

Key coefficient: - $\beta_3 < 0$ → evidence of inverted-U

Estimation

- Logistic regression OR linear probability model
- Cluster standard errors at team level

VII. Nonparametric Analysis

Before imposing functional form:

1. Bin pool quality into deciles
2. Plot promotion probability vs pool quality

Goal: Visually confirm inverted-U pattern.

VIII. Identification Concerns

1. Sorting into Teams

- Better players may join better teams

Mitigation: - Control for observable ability - Optional: recruit rankings (if available)

2. Mechanical Correlation

- Pool includes individual

Solution: - Leave-one-out mean ✔

3. Exposure vs Ability

- Strong teams provide visibility

Interpretation: - This is part of the mechanism, not purely bias

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IX. Extensions

Interaction Effects

Performance × Pool Quality

Heterogeneity

- By position
- By class year
- By team quality

Alternative Pool Definitions

- Team × season
- Coach × season
- Position group within team

X. Expected Findings

- Low-quality teams → low advancement probability
- Moderate-quality teams → highest advancement probability
- High-quality teams → declining marginal advancement probability

Interpretation: Tradeoff between signal quality and congestion.

XI. Pseudocode (Illustrative)

Construct leave-one-out pool quality

df['pool_mean'] = df.groupby(['team','year'])['perf'].transform('mean') df['pool_lo'] = (df['pool_mean']*df['n'] - df['perf']) / (df['n'] - 1)

Estimate model

import statsmodels.formula.api as smf

model = smf.logit('pro_entry ~ perf + pool_lo + I(pool_lo**2) + C(year) + C(position)', data=df).fit()

XII. Contribution

If results replicate:

> *Nonlinear peer effects in advancement are a general feature of competitive evaluation systems, not unique to the military.*

This would demonstrate that: - advancement depends on both **individual performance** and **peer context** - and that peer context operates through **both signal enhancement and congestion**

Formal Model (LaTeX)

[Y_i = _1 _i + *2* {jt}^{(-i)} + *3 (*{jt}$^{\{(-i)\}}$2 + X_i'+ _i]

[*{jt}^{(-i)} =* {k i} _{kt}]

[_3 < 0]